William Pearman
Curriculum vitae

Research summary

I am a broadly trained molecular and microbial ecologist with principal expertise in population genomics and microbial ecology. I specialize in the analysis of community data from microbial communities (either metabarcoding or metagenomics), and in eDNA analysis. My current research explores how host microbiomes influence and interact with their host along both evolutionary and ecological timescales. To undertake this research, I develop eco-evolutionary simulations of microbial communities to develop hypotheses which I then test using observational data and experiments. My research background is a combination of population genomics and microbial ecology, with my current research interests working to align these two fields to understand the evolution of host-microbe relationships. My current work combines eco-evolutionary modelling of holobiont evolution, and experimental work with Daphnia to determine microbial contributions to host fitness and to understand host-microbial responses to environmental perturbations. In addition to these areas, I am also actively involved in the application and optimisation of new molecular biology protocols for non-model organisms.

I'm also interested philosophy of science, and am currently collaborating with philosophers on the value/utility of the holobiont concept, alongside sensible null hypotheses for studying host microbiomes.

Contact

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Education

2023	**PhD Marine Science**, University of Otago, Dunedin, New Zealand
2020	**MNatSci (Distinction) Genetics**, Massey University, Auckland, New Zealand
2018	**BNatSci Genetics**, Massey University, Auckland, New Zealand

Selected teaching experience

- Contributed course material — BIOSCI 761
- Workshop instructor — SLiM eco-evolutionary simulation workshop
- Workshop instructor (upcoming) — eDNA bioinformatics workshop
- Undergraduate teaching laboratory instructor/demonstrator
- Postgraduate student supervision (multiple students)

Selected talks

2026	Invited Talk - Theoretical Microbial Ecology - ISME20 in Auckland

2026 Invited Talk - Environmental 'Omics - Queenstown Research Week

2024 Invited talk — Center for Computational Evolution's 2024 Research Showcase

2022 Conference presentation — New Zealand Molecular Ecology Conference (awarded Best PhD Presentation)

Selected talks (cont.)

2020 Conference presentation — New Zealand Molecular Ecology Conference (awarded Best PhD Presentation)

2020 Invited participant — Ira Moana Investigator Workshop

2019 Invited participant — Ira Moana Project Early Career Workshop

2017 Conference presentation — New Zealand Molecular Ecology Conference (awarded Best Student Presentation)

Selected service

2025-present Co-chair - University of Auckland Faculty of Science Research Fellow Society

2025-present Member - University of Auckland Faculty of Science Research Committee

2024 EEB Representative - University of Auckland Faculty of Science Research Fellow Society

Professional development

2026 Visiting Scholar - KU Leuven

2020 Invited participant - Ira Moana Investigator Workshop

2019 Invited participant - Ira Moana Project Early Career Workshop

Awards and honours

2022 Best PhD Presentation - New Zealand Molecular Ecology Conference

2020 Best PhD Presentation - New Zealand Molecular Ecology Conference

2020 New Zealand Marine Sciences Society Student Research Award ($1,500)

2020 University of Otago Doctoral Scholarship ($27,500)

2019 Massey University Masters Scholarship ($15,000)

Awards and honours (cont.)

2018 Natural Sciences Masters Scholarship - Massey University ($10,000)

2017 Best Student Presentation - New Zealand Molecular Ecology Conference

2015 Vice Chancellors Natural Science Excellence Scholarship, Massey University ($15,000)

Selected major grants

2025 Mana Tūāpapa Future Leader Fellowship ($820,000 NZD) - Independent 4-year fellowship to explore Daphnia microbial ecology

2025 Te Pūnaha Matatini Seed Fund ($4,880 NZD)

2025 ISME Scholar Mobility Award ($3,000 NZD)

2025 Emerging Researchers Network - Seed Fund ($2,500 NZD)

2025 Genetics Society of Australasia - Workshop Support Fund ($1,000 NZD)

2025 Center for Computational Evolution - Workshop Fund Award ($3,500 NZD)

2025 Genomics Aotearoa - Workshop Fund Award ($5,000 NZD)

2024 University of Auckland Research Fellow Society Seed Funding ($5,000)

Publications

19. Ryder, F.J., **Pearman, W.S.**, Layton, C., Parvizi, E., Johnson, C., Bellgrove, A. & Fraser, C.I. (In Press). Contrasting patterns of population structure in two habitat-forming kelp species in southeastern Australia. *Journal of Phycology*. doi:10.1111/jpy.70140

18. Kroos, G.C., Fernandes, K., Seddon, P., Ashcroft, T., **Pearman, W.S.** & Gemmell, N.J. (2026). Targeted Airborne eDNA of an Invasive Wallaby: Effects of Sampler Type, Distance, and

Environmental Conditions. *Environmental DNA*, 8(1), e70240. doi:10.1002/edn3.70240

17. Pillay, P., dos Remedios, N., **Pearman, W.S.**, Santure, A.W. & Allen, M.S. (2025). Bones, barcodes, and biodiversity: Optimising bulk bone metabarcoding analysis for tropical subfossil collections from Polynesia. *Quaternary International*, 110099. doi:10.1016/j.quaint.2025.110099

16. Pearman, W.S., Rodrigo, A.G. & Santure, A.W. (2025). Within-host microbial selection and multiple microbial generations buffer the loss of host fitness under environmental change. *FEMS Microbiology Ecology*, fiaf089. doi:10.1093/femsec/fiaf089

15. Morales, S.E., Tobias-Hünefeldt, S.P., Armstrong, E., **Pearman, W.S.** & Bogdanov, K. (2025). Marine phytoplankton impose strong selective pressures on in vitro microbiome assembly, but drift is the dominant process. *ISME Communications*, 5(1), ycaf001. doi:10.1093/ismeco/ycaf001

14. Pearman, W.S., Duffy, G.A., Smith, R.O., Currie, K.I., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Host dispersal relaxes selective pressures in rafting microbiomes and triggers successional changes. *Nature Communications*, 15(1), 10759. doi:10.1038/s41467-024-54954-z

13. Pearman, W.S., Arranz, V., Carvajal, J.I., Whibley, A., Liau, Y., Johnson, K., Gray, R., Treece, J.M., Gemmell, N.J., Liggins, L., Fraser, C.I., Jensen, E.L. & Green, N.J. (2024). A cry for kelp: Evidence for polyphenolic inhibition of Oxford Nanopore sequencing of brown algae. *Journal of Phycology*, 60(6), 1601-1610. doi:10.1111/jpy.13513

12. Pearman, W.S., Duffy, G.A., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Long-distance movement dynamics shape host microbiome richness and turnover. *FEMS Microbiology Ecology*, fiae089. doi:10.1093/femsec/fiae089

11. Pearman, W.S., Adams, C.I.M., Monteiro, M., Quesada, A. & Fraser, C.I. (2024). Fine-scale phylogenetic diversity gradients support the Antarctic geothermal refugia hypothesis. *Antarctic Science*, 1-8. doi:10.1017/S0954102024000099

10. Pearman, W.S., Morales, S.E., Vaux, F., Gemmell, N.J. & Fraser, C.I. (2024). Host population crashes disrupt the diversity of associated marine microbiomes. *Environmental Microbiology*, 26(3), e16611. doi:10.1111/1462-2920.16611

9. Pearman, W.S., Duffy, G.A., Liu, X.P., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Macroalgal microbiome biogeography is shaped by environmental drivers rather than geographical distance. *Annals of Botany*, 133(1), 169-182. doi:10.1093/aob/mcad151

8. Thompson, B., Atsawawaranunt, K., Nehmens, M.C., **Pearman, W.S.**, Perkins, E.O., Pipek, P., Rollins, L.A., Tan, H.Z., Whibley, A., Santure, A.W. & Stuart, K.C. (2024). Population Genetics and Invasion History of the European Starling Across Aotearoa New Zealand. *Molecular Ecology*, e17579. doi:10.1111/mec.17579

7. Pearman, W.S., Urban, L. & Alexander, A. (2022). Commonly used Hardy-Weinberg equilibrium filtering schemes impact population structure inferences using RADseq data. *Molecular Ecology Resources*, 22(7), 2599-2613. doi:10.1111/1755-0998.13646

6. Liu, X.P., Duffy, G.A., **Pearman, W.S.**, Pertierra, L.R. & Fraser, C.I. (2022). Meta-analysis of Antarctic phylogeography reveals strong sampling bias and critical knowledge gaps. *Ecography*, 2022(12), e06312. doi:10.1111/ecog.06312

5. Fraser, C.I., Dutoit, L., Morrison, A.K., Pardo, L.M., Smith, S.D.A., **Pearman, W.S.**, Parvizi, E., Waters, J. & Macaya, E.C. (2022). Southern Hemisphere coasts are biologically connected by frequent, long-distance rafting events. *Current Biology*, 32(14), 3154-3160.e3. doi:10.1016/j.cub.2022.05.035

4. Pearman, W.S., Wells, S.J., Dale, J., Silander, O.K. & Freed, N.E. (2022). Long-read sequencing reveals atypical mitochondrial genome structure in a New Zealand marine isopod. *Royal Society*

Open Science, 9(1), 211550. doi:10.1098/rsos.211550

3. **Pearman, W.S.**, Wells, S.J., Silander, O.K., Freed, N.E. & Dale, J. (2020). Concordant geographic and genetic structure revealed by genotyping-by-sequencing in a New Zealand marine isopod. *Ecology and Evolution*, 10(24), 13624-13639. doi:10.1002/ece3.6802

2. Arranz, V., **Pearman, W.S.**, Aguirre, J.D. & Liggins, L. (2020). MARES, a replicable pipeline and curated reference database for marine eukaryote metabarcoding. *Scientific Data*, 7(1), 209. doi:10.1038/s41597-020-0549-9

1. **Pearman, W.S.**, Freed, N.E. & Silander, O.K. (2020). Testing the advantages and disadvantages of short- and long-read eukaryotic metagenomics using simulated reads. *BMC Bioinformatics*, 21(1), 1-15. doi:10.1186/s12859-020-3528-4